

## Experiment No: 10

**Title:** Creating Database/ Table Space

- Managing Users: - Create User, Delete User
- Managing Passwords
- Managing roles: - Grant , Revoke

**Objective:**

To understand the concept of administrative commands

**Theory:**

**DATABASE** is collection of coherent data.

To create database we have :

Syntax:        CREATE DATABASE <database\_name>

Example :      CREATE DATABASE my\_db;

**TABLESPACE:**

The oracle database consists of one or more logical storage units called *tablespaces*. Each tablespace in an Oracle database consists of one or more files called *datafiles*, which are physical structures that conform to the operating system in which Oracle is running.

Syntax:

CREATE<tablespace name> DATAFILE'C:\oracle\app\oracle\product\10.2.0\server \<file name.dbf 'SIZE 50M;

Example:

Create tablespace te\_cs DATAFILE 'C:\oracle\app\oracle\product\10.2.0\server\usr.dbf 'SIZE 50M;

**CREATE USER:**

The DBA creates user by executing CREATE USER statement.

The user is someone who connects to the database if enough privilege is granted.

**Syntax:**

```
SQL> CREATE USER <username> -- (name of user to be created )  
      IDENTIFIED BY <password> -- (specifies that the user must  
                                login with this password)
```

SQL> user created

**Eg:** create user *James* identified by *bob*;

(The user does not have privilege at this time, it has to be granted. These privileges determine what user can do at database level.)

**PRIVILEGES:**

A privilege is a right to execute an SQL statement or to access another user's object.

In Oracle, there are two types of privileges

- ❖ System Privileges
- ❖ Object Privileges
- **System Privileges :** are those through which the user can manage the performance of database actions. It is normally granted by DBA to users.

Eg: Create Session, Create Table, Create user etc..

- **Object Privileges :** allow access to objects or privileges on object, i.e. tables, table columns. tables, views etc.. It includes alter, delete, insert, select update etc.

(After creating the user, DBA grant specific system privileges to user)

**GRANT:**

The DBA uses the GRANT statement to allocate system privileges to other user.

**Syntax:**

```
SQL> GRANT privilege [privilege.... ... ]  
      TO      USER ;
```

SQL> Grant succeeded

**Eg:** Grant create session, create table, create view to James;

Object privileges vary from object to object. An owner has all privilege or specific privileges on object.

```
SQL> GRANT object_priv [(column)]  
      ON  object
```

```
        TO user;  
SQL>GRANT select, insert ON emp TO James;  
SQL>GRANT select ,update (e_name,e_address)  
        ON emp TO James;
```

### **CHANGE PASSWORD:**

The DBA creates an account and initializes a password for every user. You can change password by using ALTER USER statement.

#### **Syntax:**

```
Alter USER <some user name>  
IDENTIFIED BY<New password>
```

**Eg:** ALTER USER James

```
IDENTIFIED BY sam
```

### **REVOKE:**

REVOKE statement is used to remove privileges granted to other users. The privileges you specify are revoked from the users.

#### **Syntax:**

```
REVOKE  [privilege.. ...]  
ON object  
FROM user
```

#### **Eg:**

- REVOKE create session,create table from James;
- REVOKE select ,insert

```
ON emp
```

```
FROM James
```

### **ROLE:**

A role is a named group of related privileges that can be granted to user. In other words, role is a predefined collection of privileges that are grouped together, thus privileges are easier to assign user.

```
SQL> Create role custom;
```

```
SQL> Grant create table, create view TO custom;
```

```
SQL> Grant select, insert ON emp TO custom;
```

Eg: Grant *custom* to James, Steve;

**LAB PRACTICE ASSIGNMENT:**

1. Create user and implement the following commands on relation (Emp and Dept).
2. Develop a query to grant all privileges of employees table into departments table.
3. Develop a query to grant some privileges of employees table into departments table.
4. Develop a query to revoke all privileges of employees table from departments table.
5. Develop a query to revoke some privileges of employees table from departments table.